- width & height → Sets container size.
- margin: EdgeInsets.all(10) → Adds space outside.
- padding: EdgeInsets.all(15) → Adds space inside.
- decoration → Applies background, border, shadow, and rounded corners.
- child → Contains a Text widget inside the container.

e. Why Use the Container Widget?

- Main Benefits:
  - Adds spacing with margin and padding.
  - Styles elements with borders, colors, and shadows.
  - Wraps content to control size and positioning.

Row and Column Widget

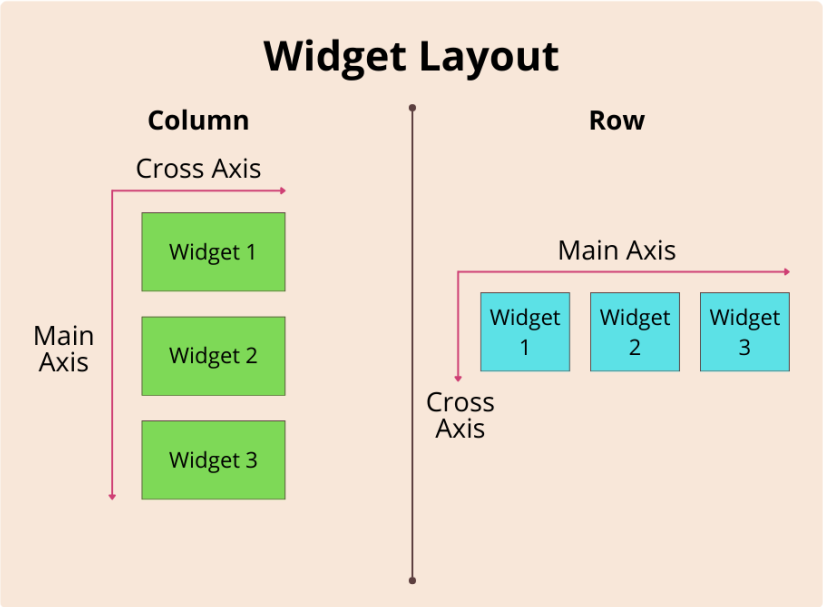
a. What are Row & Column?

- Used to arrange widgets in a horizontal or vertical layout.
- Row → Aligns widgets horizontally (side by side).
- Column → Aligns widgets vertically (one on top of another).

b. Understanding Main Axis & Cross Axis

- When using Row or Column, there are two important axes:

Axis Type	Row	Column
Main Axis	**Horizontal (左⇄右) →** Widgets are placed **side by side**.	**Vertical (上⇅下) →** Widgets are placed **one below another**.
Cross Axis	**Vertical (上⇅下) →** Controls alignment **top, center, bottom**.	**Horizontal (左⇄右) →** Controls alignment **left, center, right**.



c. Example: Row & Column in Action

- Row Example (Horizontal Layout)

```
// Row Example
Row(
  mainAxisAlignment: MainAxisAlignment.spaceEvenly, //
Aligns widgets horizontally
  crossAxisAlignment: CrossAxisAlignment.center,   //
Aligns widgets vertically
  children: [
    Icon(Icons.star, size: 40, color: Colors.blue),
    Icon(Icons.favorite, size: 40, color: Colors.red),
    Icon(Icons.thumb_up, size: 40, color: Colors.green),
  ],
),
```

- How It Works
  - `mainAxisAlignment.spaceEvenly` → Spaces icons evenly along horizontal axis.
  - `crossAxisAlignment.center` → Centers icons vertically.
  - Three icons (`Icon(Icons.star)`, etc.) are arranged side by side.
- Column Example (Vertical Layout)

```
// Column Example
Column(
  mainAxisAlignment: MainAxisAlignment.center,    // Aligns
widgets vertically
  crossAxisAlignment: CrossAxisAlignment.start,   // Aligns
widgets horizontally
  children: [
    Text('First Item', style: TextStyle(fontSize: 20)),
    Text('Second Item', style: TextStyle(fontSize: 20)),
```

```
    Text('Third Item', style: TextStyle(fontSize: 20)),
  ],
),
```

- How It Works
  - `mainAxisAlignment.center` → Centers text widgets vertically.
  - `crossAxisAlignment.start` → Aligns text to the left.
  - Three text widgets (`Text('First Item')`, etc.) appear one below another.

d. Why are Row & Column Different?

- Key Differences:

Feature	Row	Column
Layout Direction	**Horizontal (横)** → Widgets placed **side by side**.	**Vertical (縦)** → Widgets placed **one below another**.
Main Axis	**Horizontal (左⇄右)** → Controls spacing between widgets.	**Vertical (上⇅下)** → Controls spacing between widgets.
Cross Axis	**Vertical (上⇅下)** → Controls alignment (top, center, bottom).	**Horizontal (左⇄右)** → Controls alignment (left, center, right).

Button Widget

a. What are Buttons in Flutter?

- Interactive widgets that trigger actions when clicked.
- Essential for navigation, form submission, and user interaction.
- Different button types for different use cases.

b. Types of Buttons in Flutter

- Flutter provides three main types of buttons:

Button Type	Description	When to Use
TextButton	Flat button with no elevation.	Less prominent actions.
ElevatedButton	Raised button with elevation.	Highlight important actions.
OutlinedButton	Button with a border but no background.	Secondary or less critical actions.

c. Why is onPressed Important?